

Viva-Focused Short Documentation

Project: AI-Powered Restaurant Financial Management SaaS Platform

One-line idea: A MERN-based SaaS platform that unifies POS, expenses, inventory, and analytics, enhanced with AI for forecasting and expense intelligence.

1. Problem and Need

- Restaurants use disconnected tools (POS + Excel + paper receipts).
- Leads to accounting errors and no real-time visibility.
- This platform centralizes all financial workflows.

2. Objectives

- Real-time sales and expense tracking
- Accurate inventory monitoring
- Automated financial reporting
- AI-driven predictions and insights

3. System Architecture

Frontend: React dashboard

Backend: Node.js + Express APIs

Database: MongoDB

AI Layer: OCR + OpenAI + forecasting logic

4. Role-Based Access

- **Admin:** complete control and tenant management
- **Manager:** expense, inventory, analytics oversight
- **Cashier:** order and billing operations

5. Core APIs (Sample)

- `POST /api/auth/login`

- `POST /api/orders/create`
- `POST /api/expenses/add`
- `POST /api/ai/predict-sales`

6. AI Highlights

- Receipt-to-category automation
- Next-week sales prediction
- Profit/margin pattern insights
- Natural-language business assistant

7. Timeline

10-week phased plan covering architecture, module build, AI integration, testing, and deployment.

8. Viva Conclusion

This project is technically strong because it combines full-stack engineering, SaaS design, data-driven analytics, and practical AI integration for a real business domain.